



Development, methodological evaluation and application of a cell-based TRF assay for analysis of ADCC activity

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ABSTRACT

Interaction of an antibody with its FcγR plays an important role in effector functions such as antibody-dependent cell-mediated cytotoxicity (ADCC). Nowadays altered ADCC activity of an antibody can be achieved by utilizing an effective glyco-engineering strategy, which often involves changes of sugar moieties in Fc part of the antibody, thereby affecting its receptor binding with effector cells. We aimed to construct a cell-based time-resolved fluorescence (TRF) assay for the evaluation of ADCC activity triggered by the antibody drug Trastuzumab (anti-HER2) and T-DM1. The assay was initiated by incubating 2,2':6',2''-Terpyridine-6,6''-dicarboxylic acid (TDA)-labeled target SK-BR3 cells with the testing antibodies and engineered NK-92 effector cells. After incubation, the target cells were lysed to detect TDA released into the supernatant. Together with added Eu, the TDA in the supernatant formed a stable chelate of EuTDA with high-intensity fluorescence. The ADCC activity was then determined by measuring the fluorescence of EuTDA. Consequently, the method demonstrated good accuracy, precision, linearity, and specificity over methodological assessment and compared well with the Luciferase release assay in terms of the agreement of the achieved results. Using the developed assay, we evaluated the ADCC activity of two glyco-engineered anti-HER-2 antibody-drug conjugates (ADCs) and the results showed that antibody Fc glycosylation modifications influenced antibody ADCC activity to varying degrees. In conclusion, the present assay is able to accurately assess the ADCC activity induced by Trastuzumab (anti-HER2) and T-DM1, and a similar methodology can be applied to other therapeutic antibodies during drug development to help screen for antibodies with desirable ADCC activity.

1. Introduction

Since 1986 when the first murine-derived monoclonal antibodies (mAb) were created, they have been the fastest growing segment of biopharmaceuticals worldwide, with antibody drugs covering a wide range of areas such as oncology, haematological diseases, immune diseases, infectious diseases and more [1–4]. Monoclonal antibodies are immunoglobulins consisting of two light and two heavy chains linked by disulfide bonds. The Fab portion of an antibody contains the complementary determining regions which provide specific interactions with a target antigen. In contrast, the Fc portion binds to Fc receptors (FcRs) on the surface of the effector cell which triggers cellular signaling pathways and is essential in effector functions such as antibody-dependent

cell-mediated cytotoxicity (ADCC), complement-dependent cytotoxicity (CDC), and antibody-dependent cellular phagocytosis (ADCP) [5].

NK cells are the main effector cells that can exert ADCC effects. ADCC involves antibody drugs to specifically bind to antigenic epitopes on the surface of target cells, and thereafter NK cells to bind via FcγRIII receptors on their surface to the Fc segment of specific antibodies already bound to target cells, thus activating NK cells and releasing cytotoxic substances such as perforins and granzymes to kill target cells [6–8].

Antibody engineering including glyco-engineering provides a valid approach to generating improved antibody therapeutics with altered binding activities for Fc receptors. For one example, by using alanine scanning mutagenesis of individual solvent-exposed residues on the human IgG1 Fc domain, residues involved in the binding site for human

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FcR were mapped and IgG1 Fc mutant variants, namely T256A, K290A, S298A, E333A, and K334A were identified with improved binding affinity to FcγRIII and demonstrated enhanced ADCC activity *in vitro* [9]. For another example, by glyco-engineering the Fc N-glycan to reduce core fucose, the affinity between the antibody and the FcγRIII was increased, thereby also enhancing ADCC activity [10,11]. Thus, engineering of antibodies can increase the killing effect, reduce toxicity, extend the half-life to facilitate administration and reduce manufacturing costs, among other effects [12,13].

Detection of ADCC activity of antibodies is needed in antibody drug discovery and development phase to facilitate search for antibodies with desirable ADCC activity. A number of assays are currently available, involving the use of different technologies. Of note, there are advantages and disadvantages associated with each of the currently applied methods. The Chromium 51 (⁵¹Cr) release assay, the gold standard for ADCC assays, is used to detect target cell death, but this method is radioactive and associated with a complicated assay procedure [14]. In addition, the release of the intracellular enzymes such as lactate dehydrogenase (LDH) and glyceraldehyde-3-phosphate dehydrogenase (GAPDH) has also been used as an indicator of cell death, but these types of methods cannot distinguish between target and effector cells [15]. The flow cytometry-based methodologies, for example, target cells are labeled with carboxyfluorescein diacetate succinimidyl ester (CFSE) and effector cells are labeled with 7-amino-actinomycin D (7-AAD) or propidium iodide. Cytotoxicity is detected using flow cytometry. This method can distinguish target cells from effector cells with good sensitivity, but the steps are cumbersome, relatively expensive and not suitable for high throughput screening [16,17]. ADCC reporter gene assays such as the one using Jurkat T cell lines, which are used to detect signaling but do not reflect target cell death, have unknown relevance *in vivo* [18]. The luciferase reporter gene assay is performed by constructing a tumor cell line stably expressing a GFP and luciferase (GL) reporter gene, and involving the addition of antibodies and effector cells, incubation of the mixture and then addition of luciferin to react with luciferase in live cells [19]. However, the assay requires the construction of new cell lines and has the disadvantage of being cumbersome, time-consuming and expensive. EuTDA ((2,2':6',2''-terpyridine-6,6''-dicarboxylic acid) TRF (time-resolved fluorescence) is highly sensitive, has a stable fluorescence signal and is non-radioactive. Bis(ace-toxymethyl)2,2':6',2''-terpyridine-6,6''-dicarboxylate (BATDA), used in the EuTDA TRF labels target cells more gently, making it suitable for a wider range of cell lines [20].

In recent years, new developments have been made to improve the robustness and reliability of the ADCC assays. These include the use of target cells to which a reporter gene such as for green fluorescence protein or luciferase has been stably transferred, and the use of engineered NK cell lines or alternatively engineered effector cells such as the Jurkat T cell line [21,22]. In the case where primary NK cells are used in ADCC assays, the separation of primary NK cells is often required from peripheral blood mononuclear cells (PBMCs). The isolation of primary NK cells is a time-consuming process, and furthermore, the isolation of cells from different donors is subject to variability. By comparison, engineered NK cell lines provide a source for unlimited same-quality effector cell reagent, thus allowing high throughput assay to be performed and reliable results to be generated.

SK-BR3 is a human breast cancer cell line that overexpresses human epidermal growth factor receptor 2 (HER2) on the cell surface [23]. Trastuzumab (anti-HER2) is an anti-HER2 mAb, and ado-trastuzumab emtansine (T-DM1) is an ADC consisting of a HER2-targeting mAb conjugated to mertansine (DM1) via a succinimidyl-4-(N-maleimidomethyl)cyclohexane-1-carboxylate (SMCC) linker [24]. It is known that the anti-HER2 antibody can mediate ADCC between SK-BR3 cancer cells and NK effector cells [25]. In addition, HER2 antigen can be internalized upon binding with the corresponding anti-HER2 antibody, thus allowing the ADC antigen complex to gain access to SK-BR3 cancer cells and exert the therapeutic

effect of the ADC [26].

We report here the development of a cell-based TRF assay for the detection of Trastuzumab (anti-HER2) and T-DM1-induced ADCC and its related methodological evaluation and application. The assay was developed on the basis of time-resolved fluorometry as well as the use of engineered NK-92 cells and fluorescence-enhanced ligand EuTDA (Fig. 1). Firstly, the hydrophobic BATDA entered the SK-BR3 cancer cells through the passive transport of the cell membrane, and was then hydrolyzed into the hydrophilic TDA under the action of cell esterase, leading to the target cancer cells being labeled. When the antibody like Trastuzumab (anti-HER2) and T-DM1 and NK-92 cells or a lysis buffer were added together, the cancer cells were lysed and TDA was released into the supernatant, forming a stable and highly-fluorescent chelate between TDA and Eu, and the cell lysis rate was proportional to the fluorescence signal of EuTDA. Therefore, the *in vitro* ADCC activity of the antibody like Trastuzumab (anti-HER2) and T-DM1 can be detected. Subsequently, the methodological assessment of the assay with regard to accuracy, precision, linearity, specificity and a comparison of the developed assay with the Luciferase release assay was performed. Finally, the qualified assay was applied to three anti-HER2 ADCs to evaluate their ADCC activity. Two of these ADCs had undergone glycosylation engineering modifications.

2. Materials and methods

2.1. Cell Lines, Reagents, and Antibodies

SK-BR3 cells were purchased from Procell. SK-BR3 cells were cultured in Dulbecco's modified eagle medium (DMEM) (Meilunbio) supplemented with 10% FBS (Gibco) and 1% Penicillin Streptomycin (Gibco). NK-92 cell line (No-GFP-CD16, high affinity, 176V) was obtained from Brink Biologics. Alpha-minimum essential medium (α-MEM) (Gibco) supplemented with 12.5% FBS (Gibco), 12.5% Horse Serum (Gibco), 1% Penicillin Streptomycin (Gibco) and 150 U/mL Recombinant Human IL-2 (Peprotech) was used to culture NK-92 cells. Europium Solution, BATDA Reagent, and DELFIA lysis buffer were purchased from Cisbio Bioassays of PerkinElmer. D-Luciferin Sodium Salt was purchased from Yeasen. Trastuzumab (anti-HER2) was purchased from Selleck. T-DM1 was purchased from Roche. Anti-β-Gal-hIgG1 was purchased from Invivogen. ADC1 and ADC2 were glycosyl site-specific ADCs made of Trastuzumab (Herceptin, Roche) with "one-pot chemoenzymatic glycoengineering approach". ADC4 was randomly conjugated to Trastuzumab (Herceptin, Roche) [27–29].

2.2. EuTDA TRF assay

To test Trastuzumab (anti-HER2) and T-DM1 ADCC activity, a cell-based EuTDA TRF assay was developed. Briefly, SK-BR3 cells (Target cell) re-suspended in the labeling buffer (PBS+10% (v/v) FBS+ 20 mM Hepes buffer) at a concentration of 1×10^6 cells/mL were labeled with 0.167% (v/v) BATDA in a cell culture incubator containing 5% CO₂ at 37 ± 1 °C for 10 min, and then the cells were washed 3 times with Wash buffer (PBS+20 mM Hepes buffer). After the last wash, the cells were re-suspended in the medium (DMEM + 10% FBS) and the cell concentration was adjusted to approximately 2×10^5 cells/mL. Next, 50 μL of 10⁴ labeled SK-BR3 cells was mixed with 3-fold serially-diluted antibody Trastuzumab (anti-HER2) or T-DM1 in the range from 0.00226 nM to 400 nM, and 8×10^4 NK-92 cells in a well of a U-bottom 96-well plate for a total reaction volume of 200 μL (E:T = 8:1). Meanwhile, three control sample wells were set up, namely the background wells composed of 50 μL supernatant from centrifuged labeled SK-BR3 cells and 150 μL medium; spontaneous release wells composed of 50 μL labeled SK-BR3 and 150 μL medium; and maximum release wells composed of 50 μL labeled SK-BR3, 140 μL medium and 10 μL DELFIA lysis buffer. Subsequent incubation was performed in a 5% CO₂ cell culture incubator at 37 ± 1 °C for 1 h, followed by centrifugation at

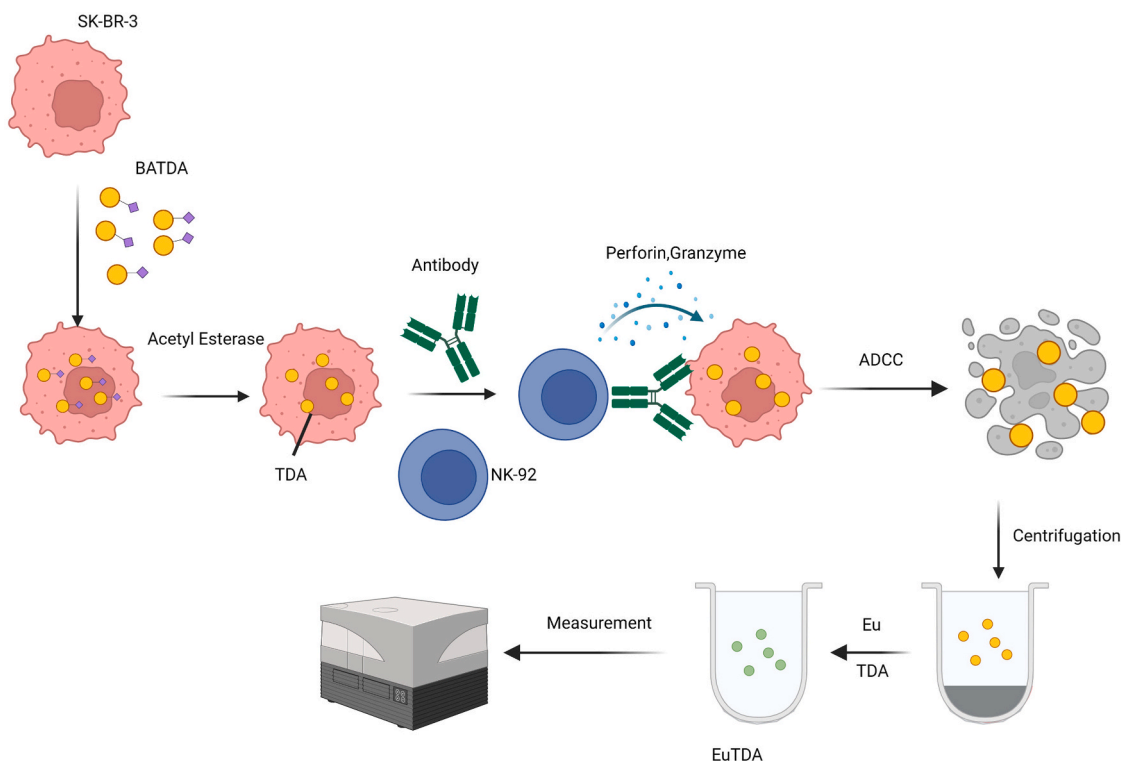


Fig. 1. Schematic representation of the EuTDA TRF assay for ADCC.

500 g for 5 min. Thereafter, 20 μ L of Eu solution was added to a HTRF 96-well low-volume plate (#66PL96025, PerkinElmer), and to each well of the plate, 2 μ L of centrifuged supernatant was transferred, followed by incubation at 250 ± 50 rpm for 15 min at room temperature. The fluorescence of the plate was measured with Spark10M microplate reader and the data was processed with SparkControl Magellan V1.2 software. A 4-parameter logistic model (GraphPad Prism 9) was used to perform curve fitting for the fluorescence values obtained at 612 nm and the log-transformed sample concentrations, generating a half maximal effective concentration (EC_{50}) value.

2.3. Methodological evaluation

According to related guidelines [30–33], the following five performance parameters were selected for methodological evaluation of the assay: intra-assay precision, inter-assay accuracy and precision, linearity, target cell robustness and specificity.

2.3.1. Intra-assay precision

To evaluate the signal variability of each antibody dilution point obtained in replicates within a batch, the testing for intra-assay precision was conducted. In practice, the diluted samples of each concentration of Trastuzumab (anti-HER2) or TDM1 were assayed in 6 replicate wells. The coefficient of variation (CV%) of fluorescence values at each concentration level should not exceed 20.0%.

2.3.2. Inter-assay accuracy, precision and linearity

To evaluate the variability and Bias% of the relative binding activity (EC_{50} ratio) values of the five tested concentrations obtained from different batches as well as the linearity between the measured relative binding activity values and the expected relative binding activity values, the testing for inter-assay accuracy, precision and linearity was conducted. Trastuzumab (anti-HER2) and T-DM1 were each set at concentrations of 600, 500, 400, 300 and 200 nM, corresponding to dilutions of 150%, 125%, 100%, 75% and 50% of 400 nM concentration, respectively, and then at least 3 independent batches were performed over a

period of 2 days at downward 3-fold serial dilutions of these concentrations. The CV% of the EC_{50} ratios obtained for the three batches of Trastuzumab (anti-HER2) or T-DM1 at different dilutions should not exceed 20.0%, and the Bias% of the mean EC_{50} ratios obtained for each dilution should not exceed 30% of the theoretical values. The average EC_{50} linear fit for the three batches should have an R^2 greater than 0.9 and a slope range of 0.7–1.3.

2.3.3. Target cell robustness

To know the extent to which the cell passage has an effect on the assay, SK-BR3 cells with cell passage number of 17, 20, 22 and 23 were chosen for robustness assessment. The CV% of the EC_{50} values obtained by using different cell generations should not exceed 20.0%.

2.3.4. Specificity

To evaluate whether the results generated by the assay are affected by the structurally related molecules but without ADCC activity, the testing for specificity was performed. In addition to Trastuzumab (anti-HER2) and TDM1, Anti- β -Gal-hIgG1 was tested as a negative control for the assay as it is of no ADCC activity. Trastuzumab (anti-HER2) and TDM1 samples should fit the 4-parameter logistic model but Anti- β -Gal-hIgG1 samples should not fit the 4-parameter logistic model.

2.3.5. Comparison of methods

SK-BR3 cells stably expressing GL were in-house made and the cell concentration was adjusted to approximately 2×10^5 /mL using a medium of DMEM supplemented with 10% FBS, and 50 μ L was added to each well of a cell culture plate. Trastuzumab (anti-HER2) was 3-fold serially diluted to multiple concentration points (from 0.00226 nM to 400 nM) and 100 μ L of 1×10^6 /mL NK-92 cells (E:T = 10:1) was added to each well of the cell culture plate. Then after incubation for 2 h, centrifugation of the plate was performed at 500 g for 5 min, and 100 μ L of supernatant was removed from each well, then 100 μ L of D-Luciferin (Sodium Salt, 0.5 mg/mL) was added to each well, followed by mixing well and incubation at 37 $^{\circ}$ C for 5–10 min. Luciferase-related luminescence values were measured using a multifunctional plate reader

(BioTek), and cytotoxicity was calculated according to the following equation: Cytotoxicity % = (1 - sample value / mean value of target cells (without Trastuzumab (anti-HER2) and NK-92 cells)) * 100. A 4-parameter logistic model (GraphPad Prism 9) was used to perform curve fitting for Cytotoxicity% values and log-transformed sample concentrations, generating individual EC₅₀.

2.4. Application of the EuTDA TRF assay

The methodologically qualified assay was applied to three antibody drug conjugates (ADC1, ADC2 and ADC4) composed of Trastuzumab (Herceptin, Roche) and MMAE (monomethyl auristatin E) [27]. ADC1 and ADC2 were prepared by a previously reported procedure “one-pot chemoenzymatic glyco-engineering approach” [28,29]. Briefly, Trastuzumab heterogeneous Fc N-glycans were released by Endo-S, then the obtained deglycosylated Trastuzumab was treated with Endo-S and Azido-tagged Sia2Gal2GlcNAc2Man3GlcNAc oxazoline (Az-SCT-ox) or Azido-tagged Gal2GlcNAc2Man3GlcNAc oxazoline (Az-CT-ox), giving glyco-engineered antibody Az-S2G2F-Trastuzumab and Az-G2F-Trastuzumab, respectively. Each glyco-engineered antibody with azido tag covalently linked with MMAE via copper-free click reaction to form two gsADCs, ADC1 and ADC2. The involving drug-linker was DBCO-Lys(PEG24)-VC-PAB-MMAE [28]. The other drug-linker SMCC-PEG4-VC-PAB-MMAE was used to conjugate to Lys residues of Trastuzumab, giving a randomly conjugated ADC4 with a DAR of about 4.

3. Results

3.1. Method evaluation

3.1.1. Intra-assay precision

Trastuzumab (anti-HER2) and T-DM1 were diluted in gradients to multiple concentration points (from 0.00226 nM to 400 nM) with 6 replicate wells for each diluted sample. We observed that the CV% of fluorescence values at each concentration level was less than 20.0% (Table 1). The results indicated that the present ADCC assay for Trastuzumab (anti-HER2) and T-DM1 has good intra-assay precision.

3.1.2. Inter-assay accuracy, precision and linearity

The CV% of the EC₅₀ ratios obtained for the three batches of Trastuzumab (anti-HER2) at different dilutions was less than 20.0%, and the Bias% of the mean EC₅₀ ratios obtained for each dilution ranged from -20.2–0.0%, all within 30% of the theoretical values, and the inter-assay precision and accuracy of Trastuzumab (anti-HER2) met the acceptance criteria (Table 2). Furthermore, the R² was 0.98 for a linear fit over the 50%–150% EC₅₀ ratio interval, with a slope of 1.04. Therefore, the Trastuzumab (anti-HER2) method assay showed linearity in the EC₅₀ ratio range of 50%–150%, equivalent to dilutions of

200–600 nM (Fig. 2A). By comparison, the CV% of the EC₅₀ ratios obtained for the three batches of T-DM1 at different dilutions was less than 20.0%, and the Bias% of the mean EC₅₀ ratios obtained for each dilution ranged from 0.0%–20.0%, all within 30% of the theoretical values, and the inter-assay precision and accuracy of T-DM1 met the acceptance criteria (Table 3). However, the mean EC₅₀ ratio obtained at 50% dilution deviated over 30% from the theoretical value and was not included in the linear fit. Notably, the related R² reached 0.95 for a linear fit over the 75%–150% EC₅₀ ratio interval, with a slope of 1.29. Therefore, the T-DM1 method assay showed linearity in the EC₅₀ ratio range of 75%–150%, equivalent to dilutions of 300–600 nM (Fig. 2B). The mean EC₅₀ of T-DM1 is 0.222 nM, which is close to the literature reported EC₅₀ value of below 0.364 nM [25].

3.1.3. Target cell robustness

We used SK-BR3 cells with cell passage number of 17, 20, 22 and 23, respectively, to run the ADCC assay of Trastuzumab (anti-HER2) and T-DM1, and the CV% of EC₅₀ values obtained from the 4-parameter logistic model fit were 17.3% and 12.3%, respectively (Table 2), which were less than 20%, indicating that the method has good robustness.

3.1.4. Specificity

It was observed that both Trastuzumab (anti-HER2) and T-DM1 showed a dose-dependent response curve. However, Anti-β-Gal-hlgG1 did not generate any specific response. We can therefore conclude that Trastuzumab (anti-HER2) and T-DM1 specifically target SK-BR3 to produce ADCC effects, whereas Anti-β-Gal-hlgG1 does not target SK-BR3 and exert no ADCC effect (Fig. 3A and B).

3.1.5. Comparison of the EuTDA TRF assay with the Luciferase release assay

The EC₅₀ values of ADCC for Trastuzumab (anti-HER2) were 0.178 nM and 0.171 nM (Fig. 4), respectively, obtained by using the EuTDA TRF assay and the Luciferase release assay, and the results were comparable.

3.2. Application of the EuTDA TRF assay

After determining the accuracy, precision, specificity, and target cell robustness of the method, the assay was applied to detect the ADCC activity of the three ADCs anti HER2 and compared with T-DM1. As seen in Fig. 5 and Table 4, the EC₅₀ of randomly coupled ADC4 and T-DM1, respectively, were comparable to each other. However, the EC₅₀ of glyco-engineered modified ADC1 and ADC2 were 1.398 nM and 0.817 nM, respectively, both greater than those of T-DM1 and ADC4. The results suggest that modifications especially on the Fc glycan moieties do affect the ADCC activity of ADCs.

4. Discussion

The EuTDA TRF assay was demonstrated to be capable of detecting the ADCC effect of Trastuzumab (anti-HER2) and T-DM1 with good precision, accuracy, specificity, linearity and target cell robustness. A similar methodology can be applied to other therapeutic antibodies to help screen for antibodies with desirable ADCC activity. During the assay, where SK-BR3 cells labeled with BATDA cannot be incubated or left waiting for long time, rather need to proceed timely to the next step of the assay. Meanwhile, NK-92 cells should be prepared in advance and added to the plate in time to avoid low signal values.

Though the mean EC₅₀ ratio at a 50% dilution for T-DM1 obtained from the three batches of experiments deviated over 30% from the theoretical value, it was once subjected to linear regression, but the obtained R² did not meet the acceptance criteria. Therefore, the mean EC₅₀ ratio at a 50% dilution point was eventually not included in the linear fit. The possible reason for this deviation is that T-DM1 is an anti-HER2 ADC consisting of Trastuzumab as the antibody component and

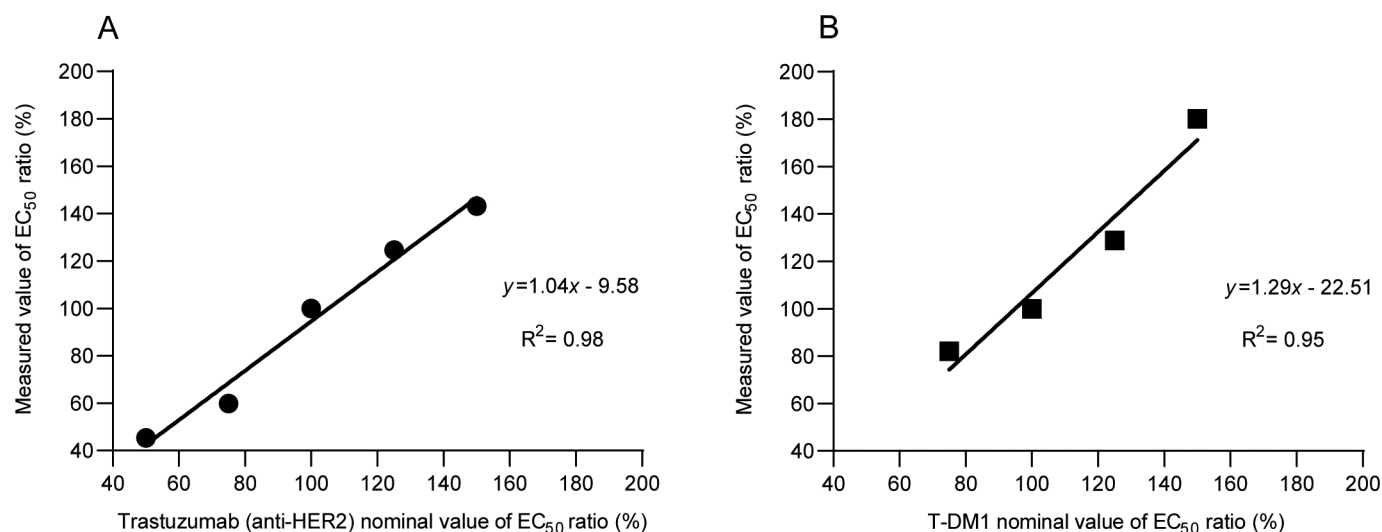
Table 1
Intra-assay precision of Trastuzumab (anti-HER2) and T-DM1.

Concentration (nM)	Trastuzumab (anti-HER2)		T-DM1	
	612 nm (N = 6)	CV (%)	612 nm (N = 6)	CV (%)
400	37229	2.1	48388	1.8
133	30449	10.6	45372	6.4
44.4	31907	7.7	41784	7.5
14.8	33636	7.9	41433	7.2
4.94	31529	5.4	39305	9.3
1.65	31739	5.2	40662	6.9
0.549	24594	9.6	35782	5.1
0.183	13466	10.7	26171	8.8
0.0610	12334	12.0	13968	11.0
0.0203	11683	15.0	13236	11.1
0.00677	10932	10.3	13256	3.8
0.00226	10488	3.0	14157	7.8

Table 2

Inter-assay accuracy, precision and linearity of Trastuzumab (anti-HER2) and T-DM1.

Antibody	Trastuzumab (anti-HER2)				T-DM1			
	Mean EC ₅₀		CV		Mean EC ₅₀		CV	
	Dilution factor	(nM) (n = 3)	(%)	Mean EC ₅₀ ratio (%)	Bias (%)	(nM) (n = 3)	(%)	Mean EC ₅₀ ratio (%)
150%		0.128	18.1	143.1	-4.6	0.123	13.4	180.1
125%		0.147	7.9	124.6	-0.3	0.172	17.6	128.8
100%		0.184	8.9	100.0	0.0	0.222	17.1	100.0
75%		0.307	5.7	59.8	-20.2	0.270	6.9	82.1
50%		0.405	4.8	45.3	-9.4	/*	/*	/*

Note: EC₅₀ ratio % = EC₅₀(100%)/EC₅₀(50–150%) × 100, * : no data.**Fig. 2.** Comparison of EC₅₀ measured values with theoretical values for Trastuzumab (anti-HER2)(A), or T-DM1(B). The X-axis represents the theoretical EC₅₀ ratio, the Y-axis represents the measured EC₅₀ ratio, and the measured EC₅₀ ratio at each point is the average value of the EC₅₀ ratios of the three batches.**Table 3**

Target cell robustness of Trastuzumab (anti-HER2) and T-DM1.

Antibody	Trastuzumab (anti-HER2)				T-DM1			
Cell passage number	17	20	22	23	17	20	22	23
EC ₅₀ (nM)	0.162	0.239	0.189	0.178	0.236	0.191	0.179	0.197
(95%CI)	0.0628–0.347	0.122–0.407	0.0934–0.343	0.119–0.276	0.0723–0.537	0.128–0.281	0.0786–0.232	0.121–0.296
CV (%)	17.3				12.3			

DM-1 as the cytotoxic component. The experimental results of DM-1, namely the cytotoxic part, constitute the extra cell killing effect of T-DM1 over those of Trastuzumab. In other words, T-DM1 requires receptor-mediated internalization to enter the lysosome, where it undergoes proteolytic degradation by lysosomal proteases, leading to the release of DM-1 attached to lysine residues and subsequent cell-killing [34]. This non-ADCC effect likely plays a minor role and is unapparent with 75–150% dilutions but it becomes relatively apparent with the 50% dilution. Thus, the mechanism of action of T-DM1 differs from that of Trastuzumab, which may explain the occurrence of results exceeding 30% at the 50% dilution point.

In comparison with the Luciferase release assay, the EuTDA TRF assay was performed with an E:T ratio of 8:1 and incubation time of 1 h. During the experiment, the Luciferase release assay adopted the same reaction conditions as the EuTDA TRF assay, but no specific signal was detected by the plate reader. When the E:T ratio of the Luciferase release assay was adjusted to 10:1 and the incubation time was adjusted to 2 h, we detected the specific signal. Of note, the EC₅₀ results for Trastuzumab (anti-HER2) were comparable for the two assays although different E:T ratios and incubation times were used. Besides, the two methods have

distinct characteristics: the Luciferase release assay requires the construction of a new cell line, involves complex steps, consumes a significant amount of time, and is expensive. Additionally, it does not maintain the native state of tumor target cells, and its applicability to other antibody drugs is a matter deserving extra investigation. On the other hand, the EuTDA TRF assay has simpler steps, shorter reaction time, relatively lower cost, and preserves the native state of tumor target cells. Based on these factors, the EuTDA TRF assay is considered superior to the Luciferase release assay.

Nowadays, novel strategies for the design of mAbs, especially with engineered Fc portions are becoming more important than ever to ensure that the therapeutic antibodies have desirable properties in terms of efficacy and safety. Glyco-engineering modifications are a common technique regarding Fc-engineering [13]. It is known that the Fc part of the IgG molecule can mediate antibody effector functions such as ADCC, CDC and ADPC. In fact, these functions are regulated by a single N-linked biphasic glycan of the heavy chain, which is located directly below the hinge region [35]. With the EuTDA TRF assay, our results on the ADCC activities of the three ADCs that differ from each other in the Fc glycosyl group demonstrate that modifications of the Fc glycosyl

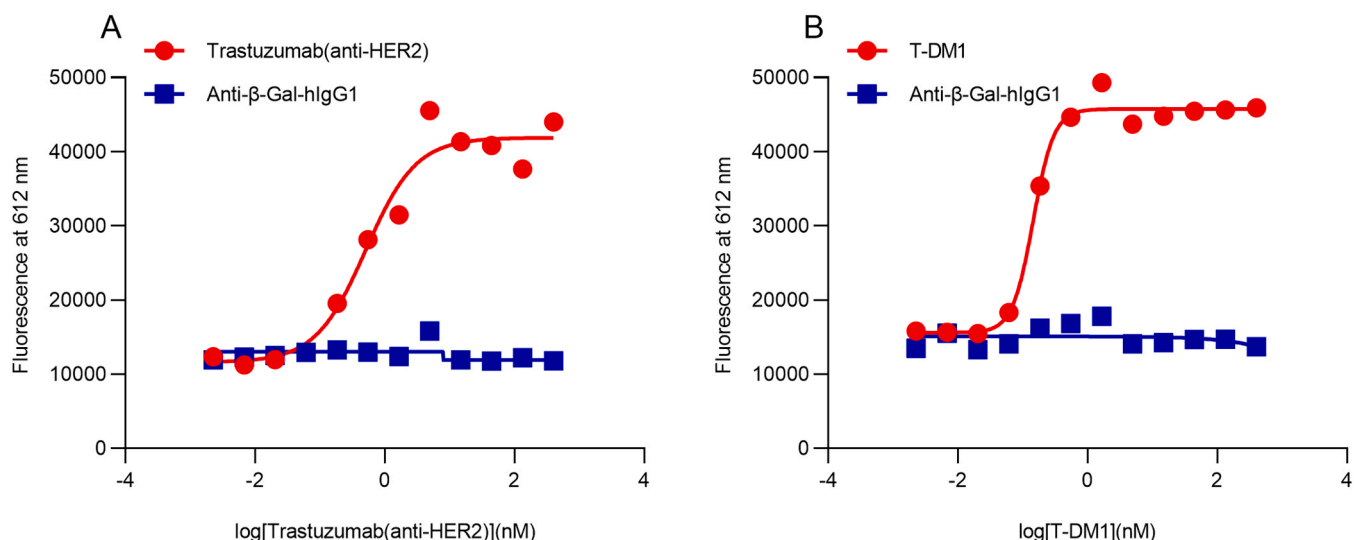


Fig. 3. Specificity test with Trastuzumab (anti-HER2) (A), T-DM1 (B) and Anti-β-Gal-hIgG1.

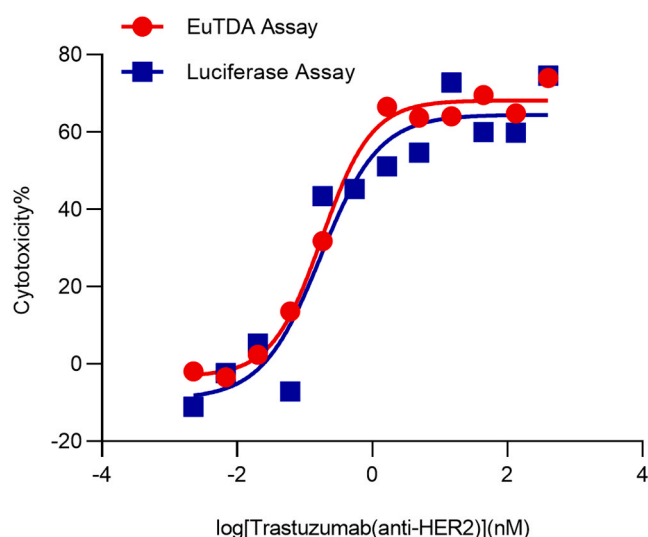


Fig. 4. Comparison of Trastuzumab (anti-HER2) using the EuTDA TRF assay versus the Luciferase release assay.

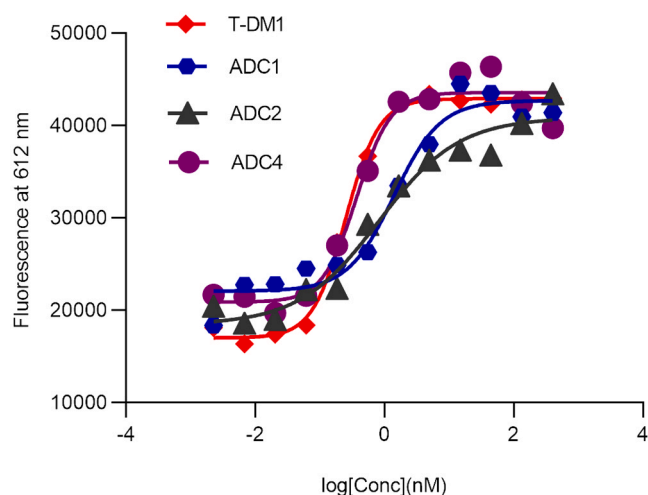


Fig. 5. ADCC of the T-DM1 and three ADCs.

Table 4

EC₅₀ values for the T-DM1 and three ADCs.

Code	Linkage	EC ₅₀ (nM)
T-DM1	Unmodified	0.250
ADC1	DBCO-Lys(PEG24)-VC-PAB-MMAE	1.398
ADC2	DBCO-Lys(PEG24)-VC-PAB-MMAE	0.817
ADC4	SMCC-PEG4-VC-PAB-MMAE	0.370

group of an antibody can affect the ADCC activity of the antibody.

5. Conclusion

The present time-resolved cell-based fluorescence assay we developed is able to accurately assess Trastuzumab (anti-HER2) and T-DM1-induced ADCC effect. Similar methodologies can be extrapolated to other therapeutic antibodies for application in the early stage of antibody drug discovery to help screen for antibodies with desirable ADCC activity.

CRediT authorship contribution statement

Xiao Zhu performed the experiments related to assay development and methodological evaluation and later on applied the developed assay three ADCs. Qiuping Qin and Likun Gong provided initial idea and supervised the whole research. Xiao Zhu and Qiuping Qin wrote the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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